

NG-KV: Necessity-Gated KV Cache Management

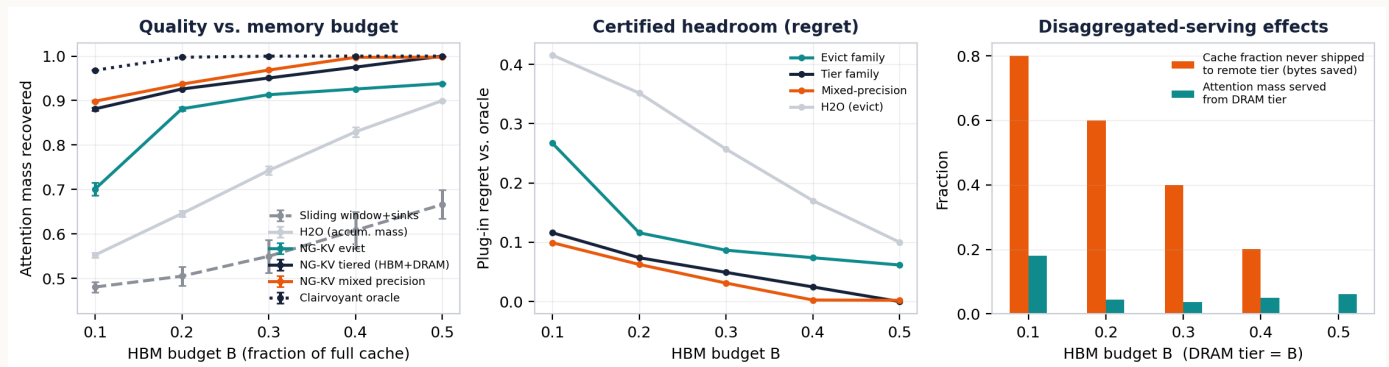
A plug-in framework for retention, tiering, and precision in LLM serving — with oracle-regret certification.

Syni open research stack · Apache 2.0 · grounded in SIVAM TR-01 (Necessity-Gated Activation Modulation) · July 2026

Thesis

Every KV-cache decision — evict or keep, HBM or host tier, 16 bits or 4 — is a necessity gate: an online estimate of how vital a cached token is to future decoding quality. LRU, sliding windows, and H2O-style accumulated-attention eviction are all implicit plug-in estimators of that necessity, with no regret story. NG-KV makes the estimator explicit (attention mass + burstiness + recency + pinned sinks), generalizes the policies it feeds (tiered placement for disaggregated serving; necessity-allocated mixed precision), and ships the evaluation methodology: replay logged traces, compute the clairvoyant oracle gate, report plug-in regret. Near-zero regret at budget B certifies near-optimality — a statement a 'quality metric didn't move' A/B gate cannot make.

Simulation results (structural traces: sinks, recency, heavy hitters, revival — 8 seeds, 2K prompt + 1K)



What moves, measurably

5x compression (B=0.20): necessity scoring recovers 0.88 of attention mass vs 0.65 for accumulated-mass (H2O-style) eviction and 0.51 for sliding windows. The gap is the revival regime: burstiness-aware necessity anticipates re-attention to dormant spans that pure history cannot.

Smooth beats hard: necessity-allocated mixed precision dominates eviction at every budget (0.997 at B=0.40 — within 0.3% of the ceiling under stated fidelity assumptions). This is the precision-of-action corollary of TR-01 made operational.

Disaggregated serving (4P2D-shaped): at B=0.20 tiering, 60% of cache blocks are never shipped to the remote tier — direct interconnect (EFA) byte savings — while the DRAM tier serves the mid-necessity mass HBM-only retention would lose.

Certified headroom: plug-in regret quantifies distance from the oracle per family and budget (evict: 0.12 at B=0.2, shrinking to 0.003 for mixed precision at B=0.4). Regret is computable offline on production traces today.

Plugin architecture: any model class, any stack

Core is framework-agnostic (NumPy; scorer is $O(T)$ state, microseconds per step) and binds via thin adapters: HuggingFace — working reference adapter for real-trace validation on any decoder (dense or MoE). vLLM — KVConnector admission/prefetch policy: score 16-token blocks (max over tokens), never ship EVICT-tier blocks, prefetch by descending necessity. SGLang — HiCache admission + eviction ranking over radix-tree nodes; LFU is the degenerate case (variance, recency weights = 0), so the generalization is strict. Production necessity signals need no attention materialization: Quest-style block key statistics, prefix-cache hit counts, MoE router entropy. Applies uniformly across GLM / DeepSeek / Kimi / Qwen-class MoE and dense models.

Stated plainly (limitations)

These are structural simulations built for relative policy ordering, not absolute production claims. Traces encode documented attention regularities but are not replays of real model attention; quantization fidelity factors are modeling assumptions from the KV-quantization literature. Validation path before any production claim: (1) real-trace replay via the HF adapter, (2) end-task quality behind gates, not mass proxies, (3) SLA-goodput measurement in the target stack. The oracle-regret methodology is unchanged at every stage.